Paleomagnetism and time-stratigraphic correlation of Proterozoic redbeds of the Labrador Trough and outliers of northern Quebec

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One hundred and forty-two oriented cores were collected from redbeds of the Sakami Formation near LG-4 (85), Dieter (19), Cambrian (18), and Snow-ball (4) lakes, all in Quebec, and from two outliers in Labrador: Evening Lake (6) and Sims Lake (10). A further 73 oriented drill cores were collected from the strongly folded Chakonipau Formation redbeds (44) in the Labrador Trough and undeformed redbeds near Lac Imbault (29) on the western edge of the trough. Thermal demagnetization of the trough redbeds yielded 52 core directions, each of which showed good specimen end points and sufficient homogeneity of magnetization, and 101 such core directions were obtained for the outlier redbeds. In most of these samples, there are no indications that the stable remanence is multi-component, and the fold test yields (for LG-4 Sakami Formation and the Chakonipau Formation) a significant (95% probability) improvement of the dispersion parameter K. Thus, the Chakonipau Formation remanence is probably pre-Hudsonian and the Sakami Formation remanence is probably pre-faulting (not dated). Furthermore, a near reversal (165°) in the Chakonipau Formation suggests that the stable remanence was acquired during or soon after deposition. The pole positions for Circum-Superior orogen rocks, the Proterozoic outliers, and the Sutton Lake inlier suggest: (1) a time-stratigraphic correlation between the LG-4 Sakami, Dieter Lake, Sutton Lake, and the upper part of the Belcher Island sequences; and (2) a time-stratigraphic correlation between the lower part of the Belcher Islands sequence and the Richmond Gulf sequence, which must be substantially older than group 1. The Circum-Superior apparent polar wander path (APWP) is drawn through magnetic North poles from the group 1 poles to the group 2 poles, yielding a slightly lower age for the trough sediments represented by the Chakonipau Formation (two opposing polarities) and the Lac Imbault redbeds. The general North American APWP is simplified by the deletion of the Richmond Gulf South poles, and seems to be positioned as much as about 30° north of the Circum-Superior APWP down to about 1800 Ma ago. This may indicate (pre-)Hudsonian movement between the northeastern part of the Canadian Shield and paleomagnetically better investigated parts to the (south-)west.

Cent quarante deux carottes orientées furent recueillies dans les couches rouges de la formation de Sakami près de LG-4 (85), des lacs Dieter (19), Cambrien (18) et Snow-ball (4) tous situés au Québec, et dans deux lambeaux d'érosion au Labrador: lac Evening (6) et lac Sims (10). En plus, 73 carottes de forage orientées furent prélevées dans les couches rouges de la formation de Chakonipau fortement plissée (44) de la fosse du Labrador et dans les couches rouges non-déformées aux environs du lac Imbault (29) sur la bordure ouest de la fosse. La désaimantation thermique des couches rouges de la fosse a fourni 52 directions et chacune exhibe de bons points finaux pour les spécimens et une homogénéité de l'aimantation suffisante, et 101 directions supplémentaires furent mesurées sur des carottes provenant des couches rouges des lambeaux d'érosion. Rien ne laisse entrevoir, pour la majorité des échantillons, que la rémanence stable est à composante multiple, et le test de plissement (appliqué à la formation de Sakami près de LG-4 et à la formation de Chakonipau) révèle une amélioration significative (95% de probabilité) du paramètre de dispersion K. Ainsi, la rémanence de la formation de Chakonipau est possiblement antéhudsonienne et la rémanence de la formation de Sakami fut acquise probablement avant l'apparition des failles. En plus, une inversion presque totale (165°) au sein de la formation de Chakonipau suggère que la rémanence fut acquise pendant ou peu après l'accumulation des matériaux. La position des pôles des roches de l'orogénèse Circum-Supérieur, des lambeaux d'érosion du Protérozoïque et de la boutonnière du lac Sutton suggèrent (1) une corrélation chronostratigraphique entre le site LG-4 de la formation de Sakami, le lac Dieter, le lac Sutton et la partie supérieure des séquences des îles Belcher, et (2) une corrélation chronostratigraphique entre la partie inférieure de la séquence des îles Belcher et la séquence du golfe de Richmond laquelle doit être considérablement plus ancienne que le groupe (1). La courbe de la dérive apparente des pôles pour l'orogénèse Circum-Supérieur passant par les pôles magnétiques nord du groupe (1) et les pôles du groupe (2) fournit un âge légèrement plus jeune pour les sédiments de la fosse représentés par la formation de Chakonipau (deux polarités opposées) et pour les couches rouges du lac Imbault. La courbe de la dérive apparente des pôles pour l'ensemble de l'Amérique du Nord est simplifiée grâce à la suppression des pôles sud du golfe de Richmond et se trouve apparemment aussi loin que 30° au nord de la courbe de la dérive apparente des pôles de l'orogénèse Circum-Supérieur pour un âge inférieur à environ 1800 Ma. Cette constatation peut indiquer un mouvement (anté-)hudsonien entre la partie nord-est du Bouclier et des parties paléomagnétiquement mieux connues au (sud-)ouest.

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Introduction

The paleomagnetism of Proterozoic rocks overlying the Archean of the Superior Province in Canada has been investigated in a band extending from Sutton Lake (Schwarz *et al.* 1982) via Richmond Gulf (Schwarz 1976; Schwarz and Fujiwara 1981) and the Belcher Islands (Schmidt 1980) to Cape Smith (Fujiwara and Schwarz 1975) into the Labrador Trough (Park 1977). The rocks sampled were largely volcanics, and the results indicate (1) relatively early Proterozoic poles,

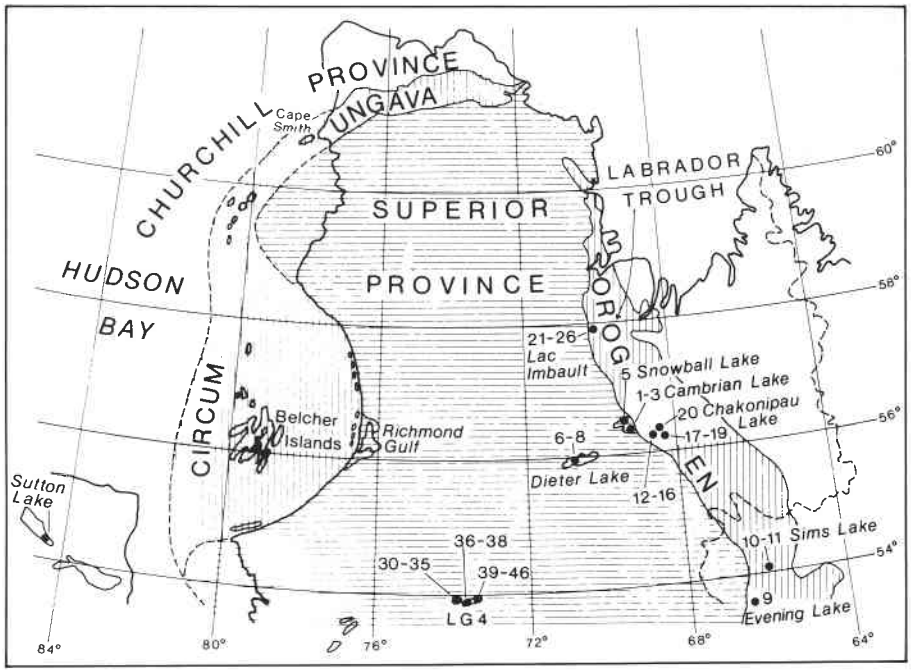


FIG. 1. General map of the Circum-Superior orogen of northern Quebec showing localities of sampling.

which form a group that does not clearly relate to poles of similar ages obtained from other areas in the Canadian Shield, and (2) younger Proterozoic poles, which appear to fall on the 1650 Ma segment of the general apparent polar wander curve relative to North America (Schwarz and Fujiwara 1981). Further results are required to compare paleomagnetically (1) the various isolated parts of the Circum-Superior Belt, and (2) the northeastern part of the Canadian Precambrian Shield with other, better investigated, parts of the shield.

Block-faulted redbeds overlying the Archean rocks in the Richmond Gulf area yielded good results comparable to those of sequences of andesitic and basaltic flows in the same area (Schwarz and Fujiwara 1981). For this reason, Proterozoic redbeds in outliers on the Superior Province of northern Quebec and Labrador were also investigated, preliminary studies having been done by Schwarz and Freda (1980) and Seguin *et al.* (1981). Furthermore, as the initial investigation by Park (1977) yielded complex results, due mostly to secondary magnetization in volcanics from the Labrador Trough, the redbed study was extended into the trough. Interpretation of the remanence directions in terms of time-stratigraphic correlations between the various pockets of redbeds should be useful in the exploration for uranium presently being conducted in these redbed occurrences.

Geology and sampling

Redbeds have been observed in the lower sedimentary parts of the Proterozoic stratigraphic columns in many areas along the western border of the Archean craton of northern Quebec (e.g., Chandler 1980; Chandler and Schwarz 1980; Wardle and Bailey 1981), as well as in fault-bounded basins on this craton (e.g., Sakami – Lac Tilly – La Forge, Dieter Lake, Cambrian Lake, Sims Lake, and Evening Lake outliers in Quebec and Labrador, and the Sutton Lake inlier in Ontario) (Fig. 1). Furthermore, they are known to occur in the Labrador Trough (e.g., Chakonipau Formation; Wardle and Bailey 1981; Clark 1979). In the outliers they are essentially flat lying and unmetamorphosed. In the Labrador Trough they occur either as flat-lying horizons (Lac Imbault) or as strongly folded sequences (Chakonipau Formation).

Redbeds are known to generally possess high stability of remanent magnetization, owing to the presence of hematite, which has made them of great interest for paleomagnetic studies. On the other hand, they are known to be amenable to the addition of stable secondary remanence components because of (re)crystallization of hematite during diagenesis and later metamorphism. It is therefore necessary (1) to compare their remanence directions with those of other rocks of similar age, or (2) to put constraints on the age of their remanence by using a fold test or by comparing statistical parameters of groups of directions in different fault blocks with different bedding orientations, or (3) preferably yet to do a conglomerate test.

For the present study, five independently oriented drill cores on average (see Table 1) were collected from each of 18 sites in the Sakami Formation (Fig. 1), three sites in the Cambrian Lake outlier, three sites in the

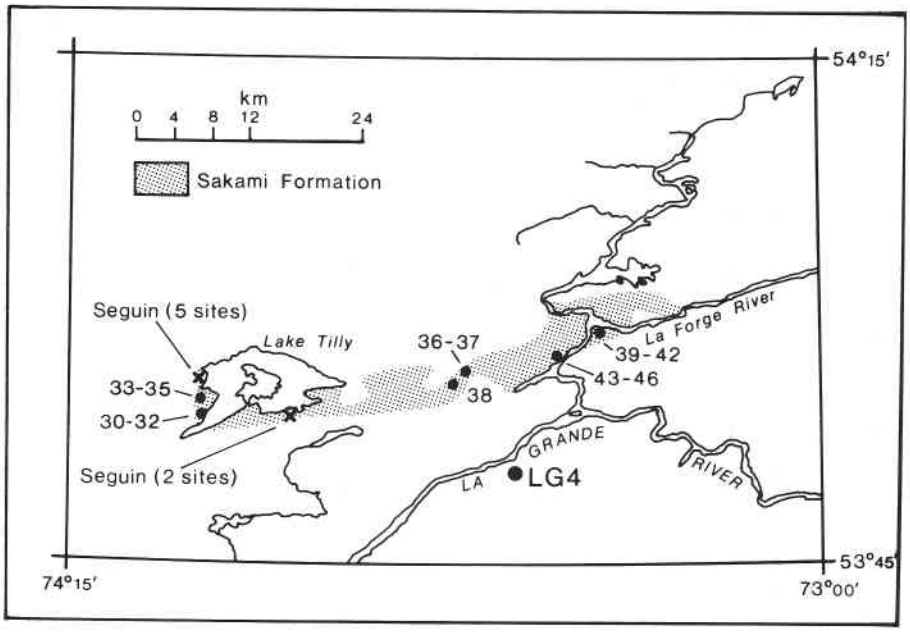


FIG. 2. Localities of sampling of the Sakami Formation near LG-4.

Dieter Lake outlier, one site each in the Snowball Lake and Evening Lake outliers, two sites in the Sims Lake outlier, and 13 sites in the Labrador Trough, yielding a total of 205 cores from 41 sites. Orientation was generally determined with a sun compass and checked with a magnetic compass. Each drill core was cut into at least two cylindrical specimens 2.5 cm in diameter by 2.2 cm long. A core direction was accepted if it had stabilized (end point) and if the angle between the specimen directions was less than 30°. The two specimen directions were averaged to obtain each core direction, which then was used to compute site directions and site statistics. These were further averaged to arrive at a direction for the formation or outlier. The results for each outlier and for the Labrador Trough specimens are discussed separately in the following pages, from east (nearest the already investigated areas of Richmond Gulf – Manitounuk Islands – Sutton Lake) to west.

Experimental method and results

The natural remanent magnetization (NRM) of each specimen was measured with a digital spinner magnetometer. At each site two specimens from different cores were subjected to demagnetization in alternating fields in steps of 50 or 100 Oe (5 or 10 mT) up to 800 Oe (80 mT), to test for stability. In general, small changes in remanence direction were observed after each step. Four specimens per site were demagnetized in detail by heating, which was found to be more effective in removing viscous components oriented roughly in the present local field direction. A stable end point was reached in about 70% of the specimens (Figs. 3, 5, 6, 8). The rest of the specimens from each site were heated in three or four steps in a selected temperature interval (normally 540–670°C) on the basis of the more detailed results obtained for the four pilot specimens. Table 1 shows that this treatment resulted in substantial improvements in the site grouping statistics, the best grouping being shown and used in the calculation of the paleopole position. Chemical demagnetization was performed on 10 samples from the Sakami Formation, but little change in intensity and direction was observed after treatment for up to 480 h, after which the attempt was abandoned. The concentration of the hydrochloric acid used was 8 *N*. The failure to show changes in the remanent magnetization may indicate that it resides in hematite or specularite grains rather than in the pigment around silicate grains. This is in accordance with the observation that the middle part of the Sakami Formation, which consists of a slightly orange coarse-grained sandstone indicating very little pigment hematite, possesses remanent intensity and stability comparable to those of the red-coloured finer grained sandstones of the upper and lower parts.

Sakami Formation

The lower sequence was sampled near the west shore of Lac Tilly (Fig. 2), where it contains a few red sandstone layers about 30 cm thick. The middle sequence is composed of orange coarse-grained sandstones east of Lac Tilley; the upper sequence contains a few red intercalations, which were sampled in two areas near the La Forge River, due north of the LG-4 dam.

TABLE 1. Paleomagnetic data for

Site	Area	NRM					Cleaned						
		N	*D*°	*I*°	K	α_{95}°	*N*	mT	°C	*D*°	*I*°	K	α_{95}°
30	SAK [low]	5	36	−42	63	10	5		620–670	41	−43	61	10
31		5	25	39	2	89	2	30		42	−41	69	31
32		5	26	−51	19	18	5		660–670	40	−53	43	12
33		6	52	−15	14	18	6		660–670	51	−29	23	14
34		6	29	−47	46	10	6		660–670	55	−55	111	6
35		6	51	5	2	56	3	30	670	70	−45	515	5
Average		6	38	−20	5	33	6			49	−45	40	11
		33	39	−26	4	15	27			49	−45	25	6
36	SAK [mid]	6	62	−57	12	20	6		660–670	90	−52	15	18
37		5	51	−60	6	35	5		660–670	85	−57	10	26
38		5	85	−33	8	30	5		660–670	95	−30	5	37
Average		3	69	−51	19	29	3			91	−46	30	23
		16	69	−51	7	15	16			90	−47	8	14
39	SAK [up]	6	9	25	4	41	6		620–650	50	−52	39	11
40		5	60	30	3	49	5		650–660	75	−55	22	17
41		3	88	−36	3	100	2		620–660	79	−53	22	56
42		4	61	62	8	36	2		620–660	61	−52	108	24
43		5	49	−2	11	24	5		620–650	55	−16	41	12
44		4	57	8	64	12	4		620–650	55	−17	13	26
45		5	39	5	10	26	5		620–660	51	−30	22	17
46		4	53	23	18	23	4		620–650	56	−20	27	18
Average		8	51	15	6	26	8			59	−37	18	14
		36	48	17	4	14	33			57	−36	13	7
Grand average	SAK	17	49	−10	4	20	17			61	−42	18	9
		85	47	−14	3	11	76			60	−42	11	5
01	CL	6	254	71	222	5	6		550	258	76	68	8
02		7	259	66	11	19	6		550	210	68	13	19
03		5	174	52	5	41	4		550	168	16	9	32
Average		3	220	69	12	37	3			189	59	4	67
		18	230	70	8	13	13			197	64	6	17
05	SNL	4	13	62	2	87	2		650	49	−63	154	20
06	DL	7	10	67	12	18	5		550	86	−58	9	27
07		6	347	69	29	13							
08		6	79	−7	11	21	5		640	78	−23	14	21
Average		3	46	52	2	108	2			81	−41	11	87
		19	44	54	3	23	10			81	−40	8	19

NOTES: SAK = Sakami Formation, [lower, middle, and upper]; CL = Cambrian Lake; SNL = Snowball Lake; DL = Dieter Lake; *N* = number of cores; *D* = Declination; *I* = Inclination of remanent magnetization; K and α_{95}: statistical grouping parameters. Pole positions are given in long., lat., and semi-axes (dm and dp) of 95% cone of confidence around the position.

Table 1 shows the site NRM directions for the three sequences. Figure 3 shows high stability of a hematite-carried remanence. Low-blocking-temperature viscous remanent magnetization (VRM) is present in a few samples, but generally the specimens' remanence decays rapidly at high temperature without much change in direction. Table 1 and Fig. 4 *A* and *B* show both a marked improvement in the grouping of the core directions and a marked difference between the directions for the lower and middle sequences. This difference is strongly reduced after the tilt correction (Fig. 4*C*), but the latter has no real effect on the grouping parameter K based on either the core directions or the site means. The lower sequence was sampled in one small area where the strike varied between 0 and 25° and the dip between 11 and 21°. The middle sequence was collected in another fairly small area where the strike varied between 74 and 99° and the dip between 20 and 38°. The ratio of the K values after and before the tilt correction calculated for the site means ($N = 3$) is 3.0, showing substantial improvement in grouping of these directions. However, this ratio is 1.0 for the core

outlier redbeds, northern Quebec

Pole				Tilt corrected				Pole			
Long.°	Lat.°	dm	dp	$D°$	$I°$	K	$\alpha_{95}°$	Long.°	Lat.°	dm	dp
				37	−54						
				28	−49						
				26	−64						
				41	−36						
				21	−64						
				63	−57						
116W	1N	14	9	37	−55	37	11	103W	5N	16	11
116W	1N	7	5	36	−55	24	6	102W	4N	8	6
				63	−61						
				52	−59						
				66	−47						
147W	22N	29	19	61	−56	88	13	120W	15N	19	14
146W	23N	18	11	61	−57	9	13	120W	15N	19	13
				33	−41						
				43	−47						
				51	−50						
				39	−43						
				49	−32						
				48	−32						
				38	−44						
				48	−36						
127W	0N	16	9	44	−41	97	6	113W	4S	7	4
126W	1S	9	5	43	−40	25	5	113W	5S	6	4
126W	4N	11	7	44	−49	37	6	111W	2N	8	5
126W	4N	6	4	44	−49	15	4	110W	2N	6	4
				82	32						
				101	31						
				158	1						
77W	6N	100	74	105	25	4	68	7W	2S	77	42
81W	13N	27	22	108	28	6	16	2W	3N	20	11
				39	−55						
				85	−44						
				72	−10						
				78	−27	10	87	143W	5N	95	52
				78	−27	8	19	142W	5N	20	11

directions (N = 16). Consequently, the fold test proposed by McElhinny (1964) and superseded by that of McFadden and Jones (1981) was not applied.

Seguin *et al.* (1981) reported a five-fold decrease in the K value for 13 independent samples collected from two small areas on the west and south shores of Tilly Lake (lower sequence only) after tilt correction. McElhinny's (1964) table indicates that the probability that the remanence is post-faulting is 99%.

The upper sequence generally contains a relatively large downward component, but after thermal demagnetization core directions correspond well with those of the lower and middle sequences (Fig. 4), but display somewhat lower negative inclinations. Within-site grouping is much improved as is the grouping of the eight site directions. Furthermore, rotating the beds to horizontal around strike yields a greater than five-fold increase in K value, indicating that the direction dispersion is significantly (probability of 0.95) different, so that the remanence is probably pre-faulting. This contradicts the result obtained by Seguin *et al.* (1981) for the lower sequence. Taking the present results for all of the

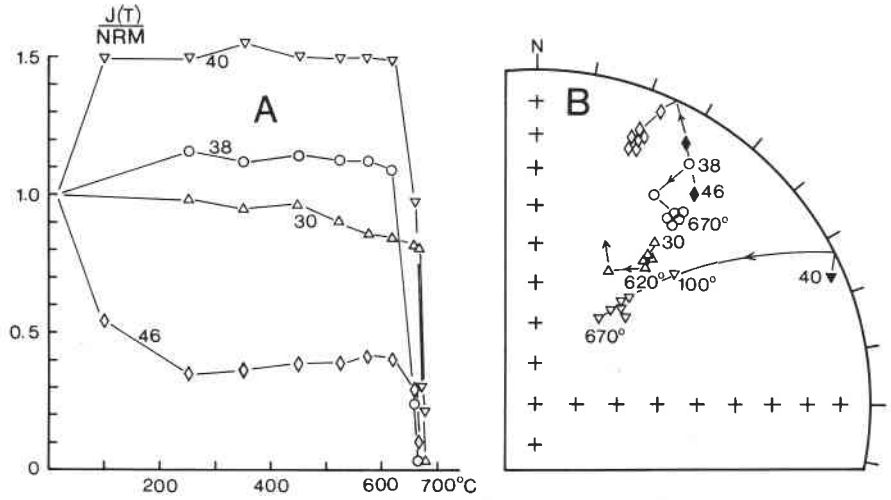


FIG. 3. Intensity decay (*A*) and direction changes (*B*) during thermal demagnetization. Open symbols indicate upward directions (negative inclination) and closed symbols represent downward directions (positive inclinations). Numbers are site/numbers (*A* and *B*) or temperatures.

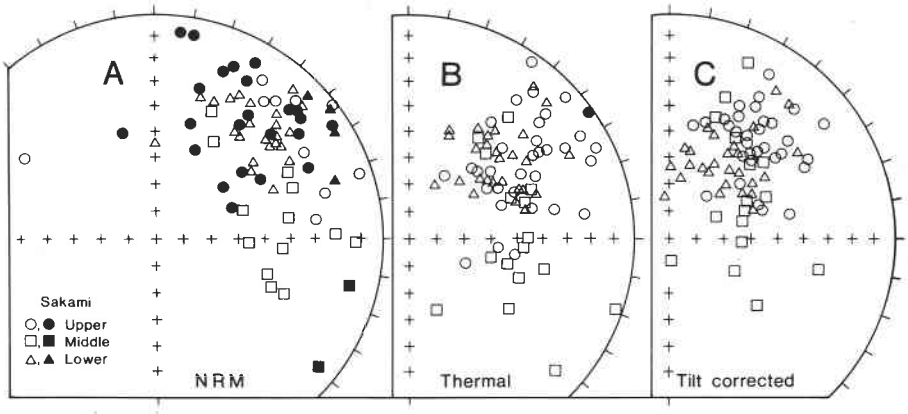


FIG. 4. Sakami Formation (outlier) core directions for NRM (*A*), accepted after thermal cleaning (*B*) and after thermal cleaning as in (*B*) followed by rotation of the beds to horizontal (*C*).

Sakami Formation, the K value is increased by a factor of two after tilt correction, again indicating with a probability of 95% that the Sakami Formation was deposited and magnetized before substantial faulting had developed. The better grouping of the core directions after tilt correction is also evident when one compares Fig. 4*B* with Fig. 4*C*.

The relative age of the magnetization with respect to the faulting is in doubt. However, there are some important differences between the results obtained by Seguin *et al.* (1981) and the results presented here.

(1) The whole formation was sampled, whereas only the lower sequence was sampled in the earlier study.

(2) The present study is based on a collection of 85 independently oriented cores collected from 17 sites distributed along the east–west band formed by the Sakami Formation (Fig. 2). The earlier study was based on 19 samples collected from seven sites in two small areas in the eastern part of the band. In the present study, the results of 76 cores were accepted out of the total of 85, whereas in the earlier study the results for 13 samples out of 19 were retained.

(3) The failure of the distribution of directions from the lower and middle sequences to show improvement in direction dispersion upon untilting may well be due to the small variation in bedding attitudes. This would cause only a small change in K, particularly since the K values are not very large and the dispersion is probably also due to other factors such as secular variation.

All accepted results, including those from Seguin *et al.* (1981), show that the stable remanence is generally the major component of the NRM and that for many speci-

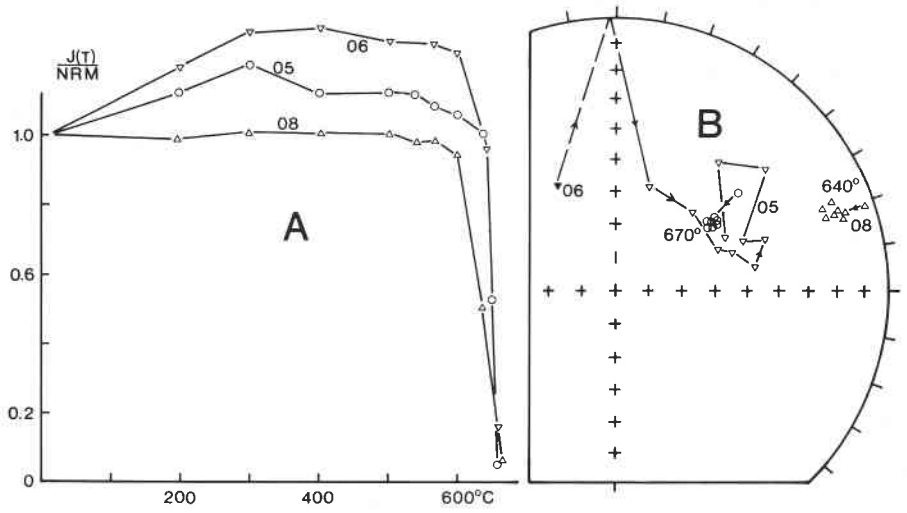


FIG. 5. Cambrian Lake and Snowball Lake outliers: intensity decay (*A*) and direction changes (*B*) during thermal treatment.

mens alternating field (AF) and thermal demagnetization have little effect on the original NRM direction. Core and sample directions define a well grouped distribution with no obvious streaking. The stable remanence is therefore probably a single-component remanence. The bulk of the conflicting evidence indicates that this remanence is pre-faulting, but the age of the faulting cannot be deduced from the regional geology.

Dieter Lake outlier

The Dieter Lake outlier could be sampled in only two locations: (1) the hilltop for which a uranium showing has been reported; and (2) a small site about 2 km farther to the west. A total of 19 cores from three sites were taken.

The results of only two of the sites were retained (Table 1). The specimens showed good end points and yielded consistent core directions. However, the site means are poorly defined, and higher core declination values are generally observed than for the Sakami Formation. Rotating the Dieter Lake beds to horizontal has no significant effect on the dispersion of the two site means, the rotation being similar in both cases. Thus, the results for the few (10) samples accepted tend to suggest that the Dieter Lake hematite remanence is roughly of the same age as that of the Sakami Formation.

Cambrian Lake outlier

Eighteen cores were collected from three sites in a stratigraphic sequence of steeply dipping redbeds about 40 m thick along the Caniapiscau River on the edge of the Labrador Trough. The bedding attitude is the same for all three sites, so that the tilt correction cannot be used to establish the age of magnetization.

Table 1 shows that the magnetization is southwest down and of high stability, the site directions only showing minor change after thermal demagnetization. This direction resembles the secondary direction reported by Fujiwara and Schwarz (1975) from Cape Smith, by Schmidt (1980) from the Belcher Islands, by Schwarz and Fujiwara (1981) from Richmond Gulf, and by Parks (1977) from areas in the Labrador Trough, all forming part of the Circum-Ungava Belt. After rotation of the beds to horizontal, a direction is found that is not known from elsewhere in the Circum-Ungava Belt. Thus, the Cambrian Lake redbeds probably were remagnetized after the Hudsonian Orogeny, which caused the main deformation in the belt. These results will not be discussed further.

Snowball Lake

The Snowball Lake outlier could only be sampled in one small outcrop. Of the four cores, only two yield good internal consistency and end points upon thermal demagnetization. These core directions (Table 1, Fig. 5) resemble the northeast-up directions of the Sakami Formation, the Flaherty Formation of the Belcher Islands (Schmidt 1980), and the Sutton Lake inlier (Schwarz *et al.* 1982). A time-stratigraphic correlation between these areas is thus suggested, but the results from the Snowball outlier are too few to be further used for the determination of a pole position.

Evening Lake and Sims Lake

A total of 16 cores at three sites were collected from the Sims Formation (Wardle and Bailey 1981). About 80% of the specimen results show no stable end points during AF and thermal demagnetization. The angle between the specimen directions defining each core direction exceeded 30° for 13 of the cores. The

TABLE 2. Paleomagnetic data for

		NRM					Cleaned					
Site	Area	N	$D°$	$I°$	K	$\alpha_{95}°$	N	°C	$D°$	$I°$	K	$\alpha_{95}°$
21	I	7	339	74	15	16	5	600	12	−30	61	10
22		6	357	24	5	34	5	600	5	−41	52	11
23		6	355	66	62	9	4	600	12	−29	16	24
24		5	354	53	72	8	6	600	5	−21	15	18
26		5	352	68	123	7	5	600	2	−16	7	32
Average		5	353	58	16	20	5		7	−28	60	10
							25		7	−28	15	8
12	CH	8	37	43	3	43	5	650	36	12	22	17
13		4	238	48	3	59	2	640	211	1	170	19
15		3	16	57	34	21	2	650	50	−27	6	137
16		7	4	45	3	43	3	650	25	−42	27	24
17		3	228	6	56	17	3	650	227	−3	128	11
18		5	203	75	2	73	3	650	199	37	15	33
19		6	209	67	15	18	2	650	212	42	36	44
20		8	238	75	91	6	7	550	214	65	28	12
Average		8	274	79	4	34	3		38	−19	7	49
							10		36	−12	6	20
							5		213	29	7	30
							17		214	29	7	16

NOTES: I = Lac Imbault; CH = Chakonipau Formation. Symbols are as used in Table 1. Average directions were computed both on the basis of site means and of core means.

remaining three cores show stable directions generally in the northeastern quadrant of the stereographic projection, but are too few and too dispersed for further analysis.

Labrador Trough

Lac Imbault

Twenty-nine cores from five sites were collected on the east shore of Lac Imbault (Fig. 7), where redbeds occur in a sedimentary sequence. Total thickness sampled was 40 m. These redbeds overlie the Archean of the Superior Province, are essentially undisturbed, and form the western edge of the Labrador Trough.

Thermal demagnetization was far more effective than AF demagnetization up to 80 mT in removing rather large viscous components with blocking temperatures up to about 300°C (Fig. 6*A*). The specimens of 25 cores showed good end points (see, for example, Fig. 6*B*) and sufficient core homogeneity to be accepted. These core directions are well grouped (Table 2) and define a north-up direction with lower declination (by about 30°) and lower inclination (by about 20°) than the Sakami Formation direction. The detailed demagnetization data (e.g., Fig. 6*B*) and the absence of streaking in the grouping of the core directions indicate that the isolated high-blocking-temperature remanence is single component.

Rotating the beds to horizontal about strike shows a marked decrease in the K value for the average direction of the site means ($N = 5$) but no change for the average based on the core directions ($N = 25$), both averages being identical (Table 2). For the middle part of the Sakami Formation, untilting also resulted in a substantial change (increase) in K based on the site means and virtually no change in K based on the core directions. It would appear that the population formed by the site means ($N = 5$ and $N = 3$) is too small to be used in this analysis. The maximum variation in bedding strike and dip for the Lac Imbault sites is 30 and 2°, respectively, so that only a very small effect on K can be expected from untilting, regardless of the effect of other sources of dispersion.

Chakonipau Formation

Forty-four cores were collected at eight sites from strongly folded Chakonipau Formation redbeds (Fig. 7). The core NRM directions show high dispersion for most of the sites (Table 2). Heating was more effective than AF demagnetization in breaking down the more stable part of the remanence (Fig. 8*A*). For most of the four pilot specimens per site, a large low-blocking-temperature component was removed (Fig. 8*B*). Twenty-seven core directions were accepted on the basis of good specimen end points and homogeneous magnetization. These fall into two groups (Fig. 9*A*, Table 2), which show high dispersion. The average direction for each of the groups calculated on the basis of the site means ($N = 3$ and $N =$

Labrador Trough redbeds

Pole				Tilt corrected				Pole			
Long.°	Lat.°	dm	dp	$D°$	$I°$	K	$\alpha_{95}°$	Long.°	Lat.°	dm	dp
				5	−24						
				357	−55						
				8	−24						
				0	−22						
				358	−25						
77W	17S	11	6	2	−30	32	14	72W	16S	15	8
77W	17S	8	5	2	−30	15	8	72W	16S	9	5
				28	−34						
				196	39						
				36	−45						
				359	−48						
				230	30						
				219	32						
				231	31						
				206	39						
107W	17S	51	29	22	−43	27	24	88W	6S	30	19
107W	21S	21	17	22	−41	14	13	89W	8S	16	10
79E	13N	33	18	217	+35	38	13	76E	8N	15	8
78E	13N	18	10	217	+35	21	9	76E	8N	11	6

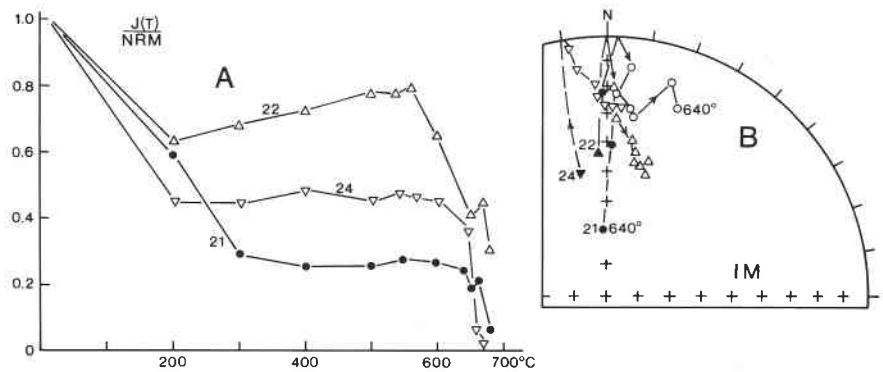


FIG. 6. Labrador Trough: Lac Imbault intensity (*A*) and direction (*B*) changes during thermal demagnetization.

5) agrees well with that calculated on the basis of the core directions ($N = 10$ and $N = 17$). The same is the case for the direction dispersion.

The direction dispersion is strongly reduced after rotating the beds to horizontal. The ratio of the K values after and before the rotation is between 2 and 5 for the four distributions considered in Table 2. In three of the four cases, the change in direction dispersion is significant on a better than 95% confidence level, whereas in the fourth case, where $N = 3$ only, the ratio is nearly 4 against McElhinny's (1964) value of 6.4. McFadden and Jones' (1981) fold test yields a positive result for this case (using their eq. [8], $f = 2.43$ and $F(4, 14) = 3.11$, for a probability level of 95%). They have pointed out that the McElhinny test is overly stringent so that a positive result obtained with this test will be supported by a positive result obtained with the McFadden–Jones test. The stable part of the remanence is therefore taken to be older than the folding or to be pre-Hudsonian. Figure 9*B* shows the improvement in grouping of the core directions and also a tendency for streaking. This streaking is roughly perpendicular to the main direction of movement of the magnetic vectors during demagnetization (e.g., Fig. 8*B*). Whether or not it

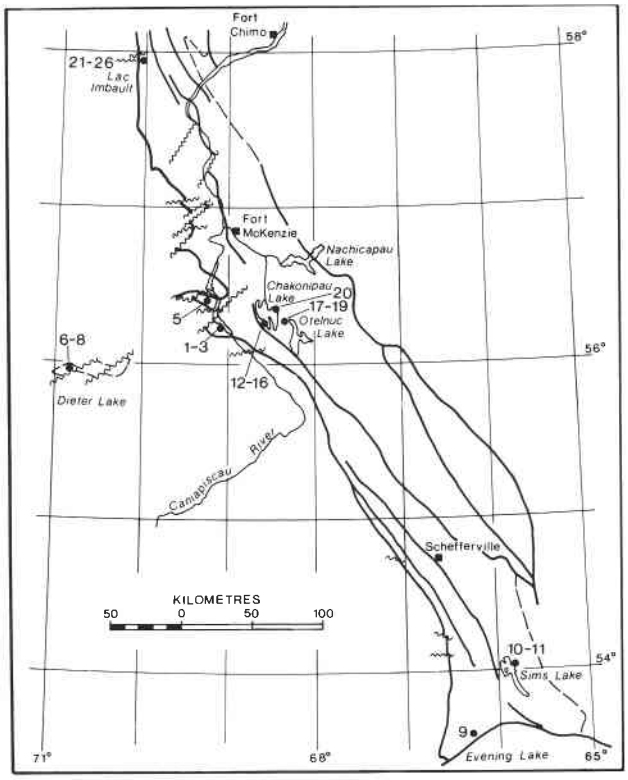


FIG. 7. Location of sampling in the Labrador Trough.

is due to secular variation or apparent polar wander in the time elapsed between deposition of the sampled beds cannot be ascertained, because the data points are few and within-site dispersion is substantial. The stratigraphic positions of the sampling sites in the 1500–3000 m thick Chakonipau Formation (Wardle and Bailey 1981) is unknown.

The average directions of the northerly-up group and the southerly-down group differ by about 165°. This indicates the occurrence of at least one geomagnetic reversal in the time that the beds were deposited (if the remanence is original), or the magnetization was acquired later (if so, probably by crystallization of hematite). If the latter mechanism has been operative, earlier magnetization (original?) must have been destroyed, as there is no good evidence that the stable remanence is multicomponent. Furthermore, sites 12 and 13, collected on the same outcrop from beds that are stratigraphically about 15 m apart, have different polarities. The bulk of the evidence shows that the magnetization was acquired shortly after deposition during diagenesis rather than during a general remagnetization period.

Pole positions

The pole positions computed from the tilt-corrected average directions of magnetization for the Sakami and Chakonipau formations and the Lac Imbault redbeds are listed in Tables 1 and 2 and shown in Fig. 10, together with pole positions determined earlier for rocks belonging to the Circum-Ungava Belt. The Sutton Lake pole is based on AF-demagnetized data (Schwarz *et al.* 1982). The Eskimo overprint (EO; Schmidt 1980) and Cape Smith pole (Fujiwara and Schwarz 1975) as well as the Chakonipau (2) pole (CH2; sites 13, 17, 18, 19, 20 in Table 2) are magnetic South poles, as are the poles from Richmond Gulf (N, O, PS, PN; Schwarz and Fujiwara 1981) and the Eskimo volcanics (EV) from the Belcher Islands (Schmidt 1980).

The following observations can be made on the basis of the results shown in Fig. 10.

(1) The results for crystalline rocks (FV, HI, SL, PS–PN, N) correspond to those for the redbeds in the same areas. The results form two groups located west and east of the South American land mass.

(2) Age relationships based on the stratigraphic sequences are known only for Richmond Gulf and the Belcher Islands. The Eskimo volcanics (EV), and the stratigraphically higher Haigh intrusives (HI) and Flaherty volcanics (FV) of the Belcher Islands all yield a positive fold test showing that their stable remanence is pre-Hudsonian, with the exception of those sites showing the post-Hudsonian Eskimo overprint EO (Schmidt 1980). The EV pole, falling in the Richmond Gulf group of poles in the southeast, is probably substantially older than the FV and HI poles, which fall near the outlier redbeds poles (DL, SAK) and the Sutton Lake inlier.

(3) The dashed apparent polar wander path (APWP) indicated by the northern Quebec–Circum Superior orogen pole positions in Fig. 10 and the relative ages for rocks for which they were determined do not correspond to the APWP drawn by McGlynn and Irving (1981), which includes a group of Nipissing poles and poles from the northwestern shield (band in Fig. 10). According to their curve, the SAK, FV, HI, and SL poles would be the oldest at 1900 Ma and the Richmond Gulf poles and EV (Belcher Islands) would be youngest at about 1775 Ma, which, on account of the stratigraphy of the Belcher Islands, means that the Richmond Gulf and EV remanence is secondary. However, the available information does not corroborate this. (a) The poles from Richmond Gulf rocks were derived from two different lava sequences and from redbeds. Although generally more complicated than the essentially single-component remanence of the Sakami Formation, good end points were observed for about 50% of the cores, and the three populations of directions are not streaked (Schwarz and Fujiwara 1981). (b) Although not conclusive on the 95% confidence level using McElhinny's (1964) tables for the dispersion parameter K, the redbed directions yielding pole Q and the andesite directions yielding

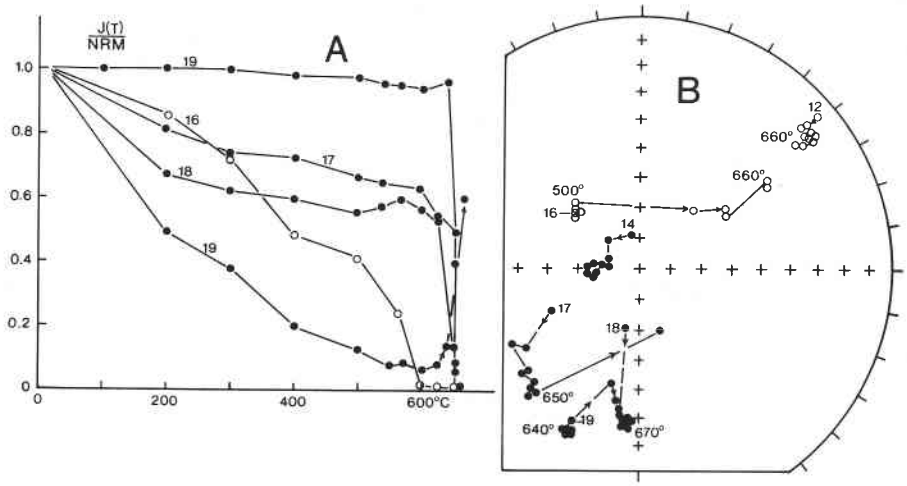


FIG. 8. Labrador Trough: Chakonipau Formation intensity decay (*A*) and direction changes (*B*).

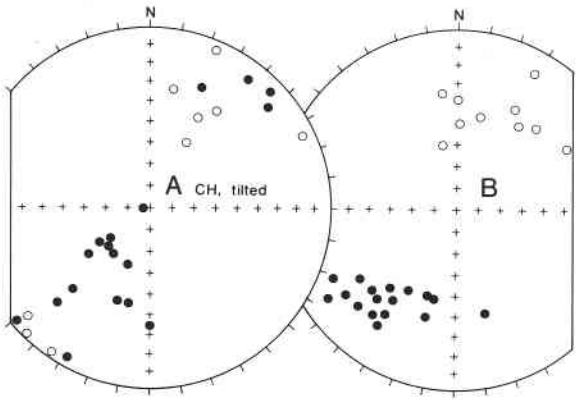


FIG. 9. Accepted core directions after thermal treatment (*A*) and the same for horizontal beds (*B*). Chakonipau Formation.

poles PS and PN show somewhat better grouping after rotating the beds to horizontal. The basalt directions yielding pole N showed no change, which conforms with the geological field evidence indicating that the faults affecting the redbeds and andesites are older than the basalts.

(4) Another way to fit the present results onto the McGlynn–Irving curve is to consider the SAK, SL, FV, and HI poles as younger than the 1750 Ma age to be assigned to the Richmond Gulf poles according to their curve, so that FV, HI, SAK, and SL would be of similar age to CS and EO. However, this is not a reasonable proposition either. (a) FV and HI poles are older than the Hudsonian Orogeny (±1750 Ma), based on a positive fold test (Schmidt 1980). (b) EO and CS are post-Hudsonian because their positions differ by only 10° before tectonic corrections, which are strongly different for the Belcher Islands (open folding, strike north–south) and the vertical bedding planes of the Cape Smith lavas (strike northeast–southwest). A similar pole is to be expected from the preliminary work on Ottawa Islands volcanics situated roughly between Cape Smith and the Belcher Islands (K. L. Buchan, personal communication, 1982).

(5) Whatever the course of the APWP, there must be a substantial difference in age between the Richmond Gulf rocks (on the western edge of the Superior craton) and the Proterozoic pockets of Dieter Lake, Sakami (La Grande), and Sutton Lake. Taking 3°/Ma as a general figure of apparent polar wander would indicate a minimum age difference of roughly 60 Ma. The Lac Imbault pole (I) is from undeformed redbeds on the eastern edge of the Superior craton. The maximum stratigraphic separation of these redbeds is about 40 m, and it is unlikely that the data are free from the effect of secular variation. The Chakonipau Formation (poles CH1 and CH2) is 1500–3000 m thick (Wardle and Bailey 1981), and the eight sampling sites probably cover a large part of this sequence, so that secular variation effects on the pole positions would be small. This formation appears to be at least 10 Ma younger than the outlier redbeds, is strongly folded, and is considered to be the oldest Proterozoic formation in the northern part of the Labrador Trough (Wardle and Bailey 1981). This suggests that sedimentation started some 70 Ma earlier on the present western edge than on the eastern edge of the Superior craton.

(6) Inverting the polarity of the Richmond Gulf and EV poles also simplifies the general APWP as drawn by McGlynn and Irving (1981). The location of these poles required this APWP to be drawn across South America to southern Africa before bending sharply northwestwards (Fig. 10). The only other relevant pole in the southern Atlantic is the Pearson Formation D pole, but the age of this pole is uncertain (compare PD in McGlynn and Irving (1978), and in Irving (1979)). If

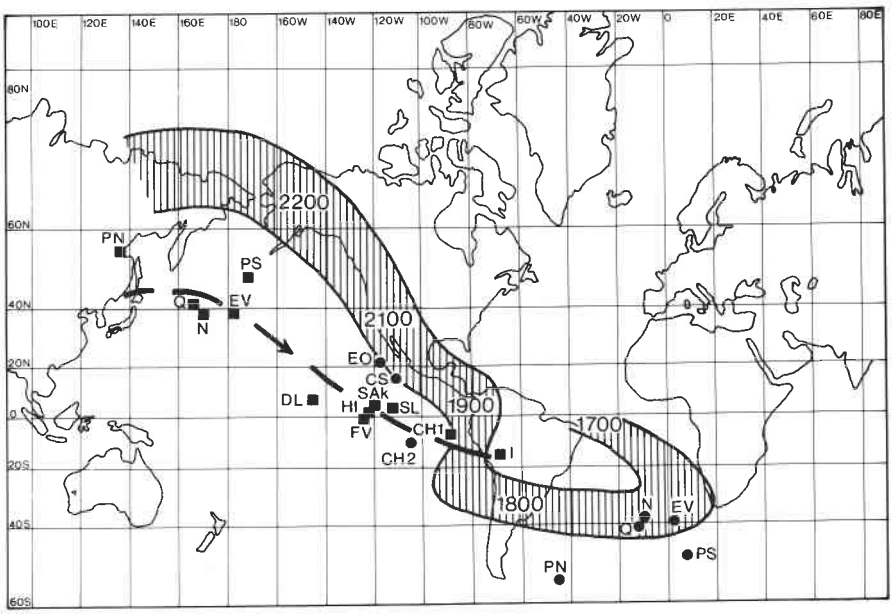


FIG. 10. Pole positions for the Circum-Superior orogen of northern Quebec. DL = Dieter Lake; SAK = Sakami Formation; SL = Sutton Lake inlier; CH1 and CH2 = Chakonipau Formation reversals; I = Lac Imbault. Richmond Gulf poles: N = Nastapoka basalt; PS, PN = Packi andesite, Q = Quingaluk redbeds. Belcher Islands; EV = Eskimo volcanics, FV = Flaherty volcanics; HI = Haig intrusives. EO and CS are Hudsonian overprints. Squares denote magnetic North poles and dots magnetic South poles. Dashed curve indicates most likely APWP derived from Circum-Superior rocks and the band indicates the APWP according to McGlynn and Irving (1981).

substantiated by further results, the difference of about 30° between the 2.1–1.8 Ma segments of the North American APWP and the Circum-Superior APWP may indicate (pre-)Hudsonian movement of the northeastern part of the Canadian Shield relative to paleomagnetically better investigated parts to the (south)west.

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